Online Retail II

Comprehensive EDA Conclusions Report

UCI E-Commerce Dataset | Dec 2009 - Dec 2010

Total Revenue	Customers	Products	Transactions	Avg Transaction
£4,506,747	**4,140**	**3,644**	**17,216**	**£261.78**

1. Data Quality & Cleaning

The raw dataset contained 525,461 records with significant noise requiring a multi-step cleaning pipeline.

Issue	Count	% of Raw
Negative/Zero Prices	3,690	0.7%
Negative/Zero Quantities	12,326	2.3%
Missing Customer IDs	107,927	20.5%
Missing Descriptions	2,928	0.6%
IQR Outliers (Price + Qty)	59,051	11.2%
Total Removed	**176,848**	**33.6%**

Final clean dataset: 348,613 records (66.4% retention rate). Outlier removal via 1.5x IQR justified - extreme values likely represent returns or bulk wholesale orders.

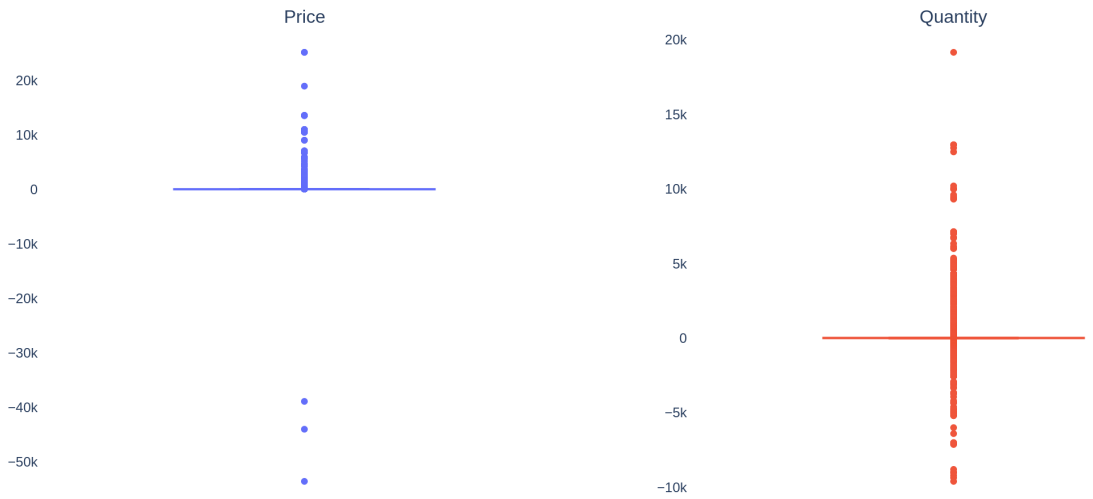

Figure 1: Box Plot showing Price and Quantity distributions before outlier removal

Conclusion: Dataset is reliable after cleaning. The box plots reveal extreme outliers in both Price and Quantity, justifying the IQR-based removal approach.

2. Product Performance Analysis

Analysis of 3,644 unique SKUs reveals extreme concentration in product performance.

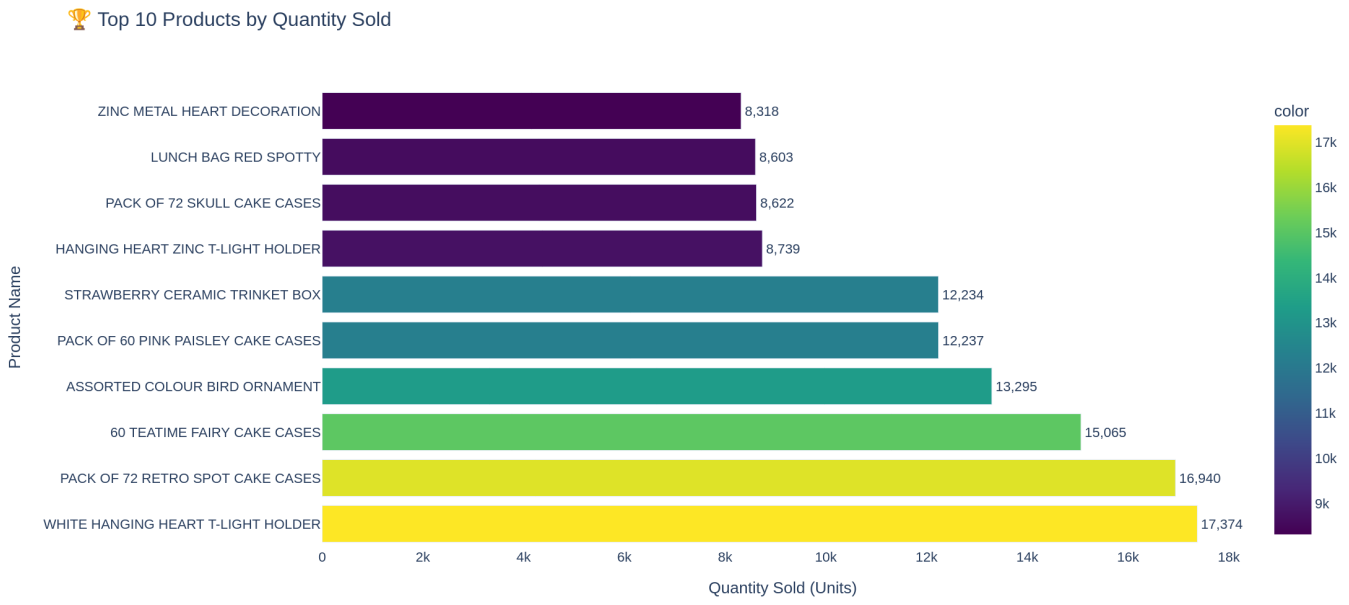

Figure 2: Top 10 products by total units sold

Conclusion: A small set of products drives the majority of sales volume. These high-volume items are predominantly low-priced decorative and gift items in the 1-5 pound range.

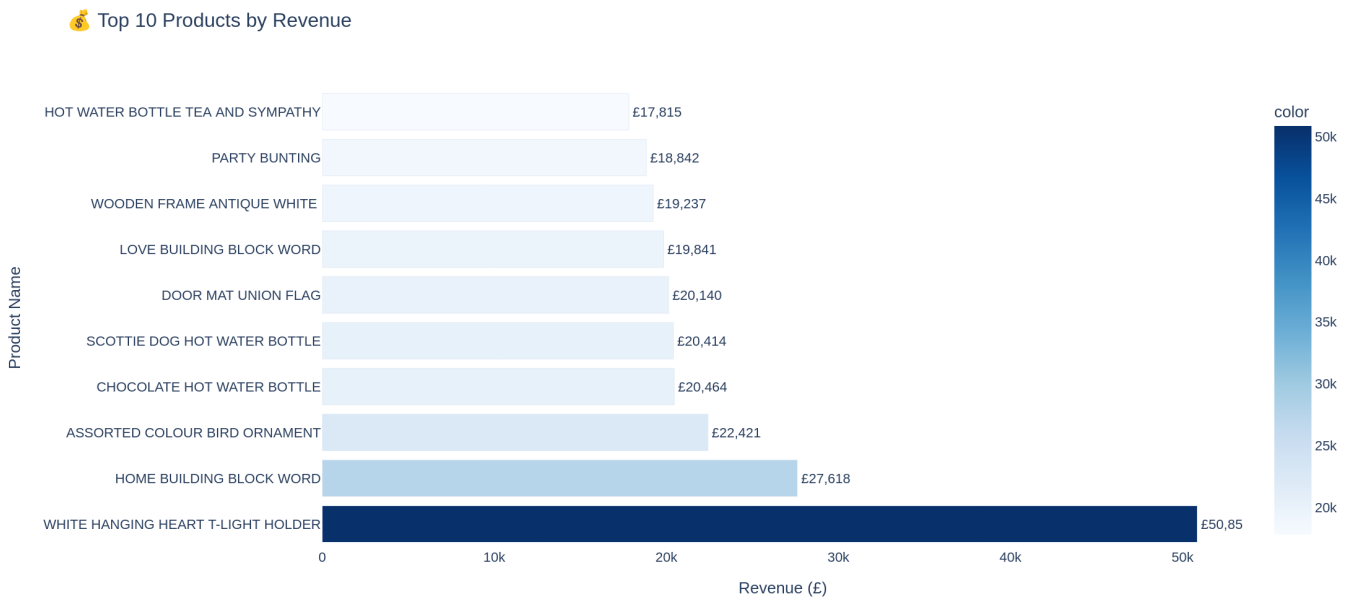

Figure 3: Top 10 products by total revenue generated

Conclusion: Revenue leaders differ from volume leaders, indicating that mid-priced items with strong volume achieve the highest total revenue. Portfolio strategy should focus on these balanced performers.

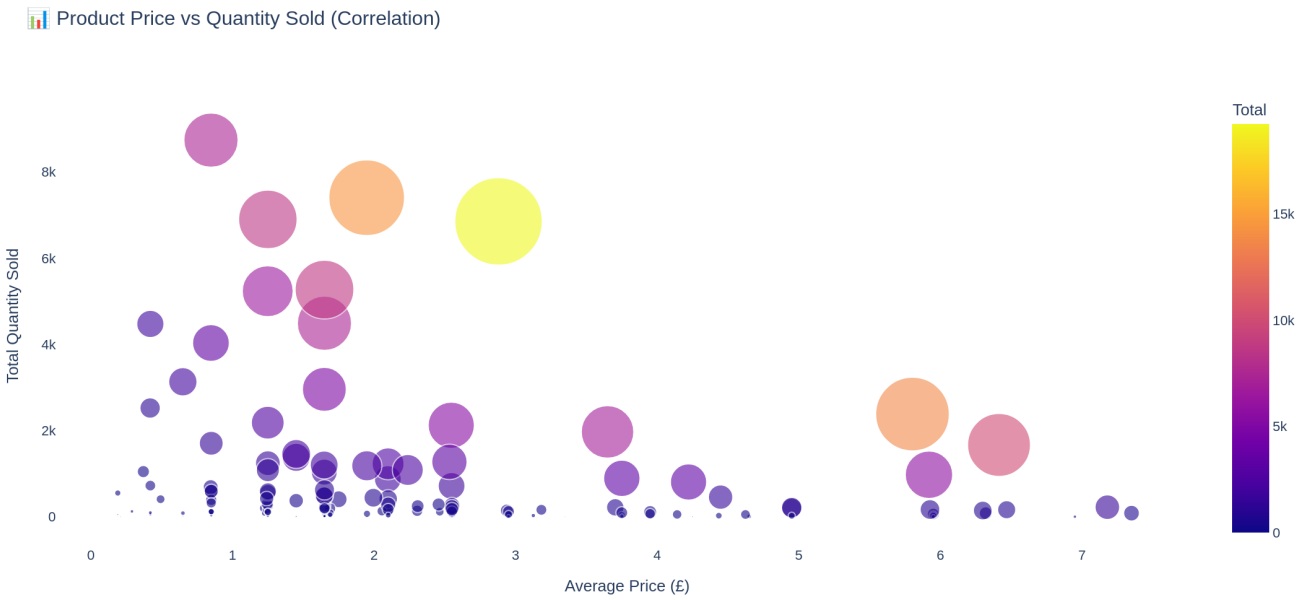

Figure 4: Price vs Quantity correlation - bubble size represents total revenue

Conclusion: Clear inverse relationship between price and quantity. Lower-priced items (under 5 pounds) sell at 10x the volume of premium items. However, the largest revenue bubbles appear at moderate price points.

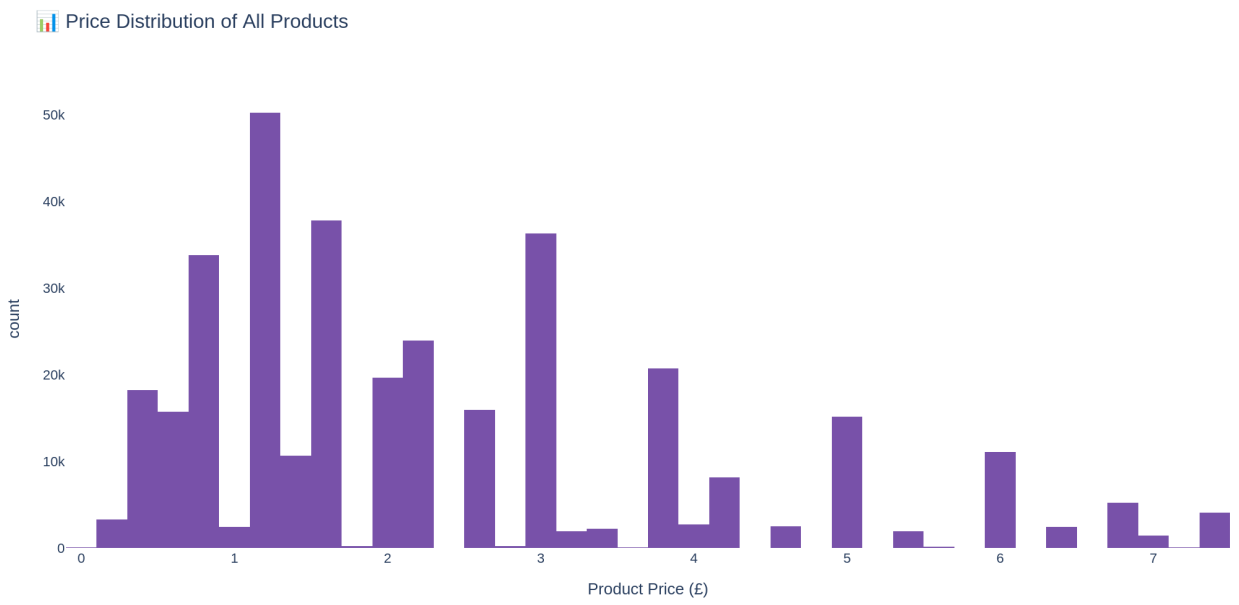

Figure 5: Price distribution histogram across all products

Conclusion: Heavily left-skewed distribution with most products clustered at 0-10 pounds. Very few premium items above 20 pounds. This confirms a volume-driven, low-price business model.

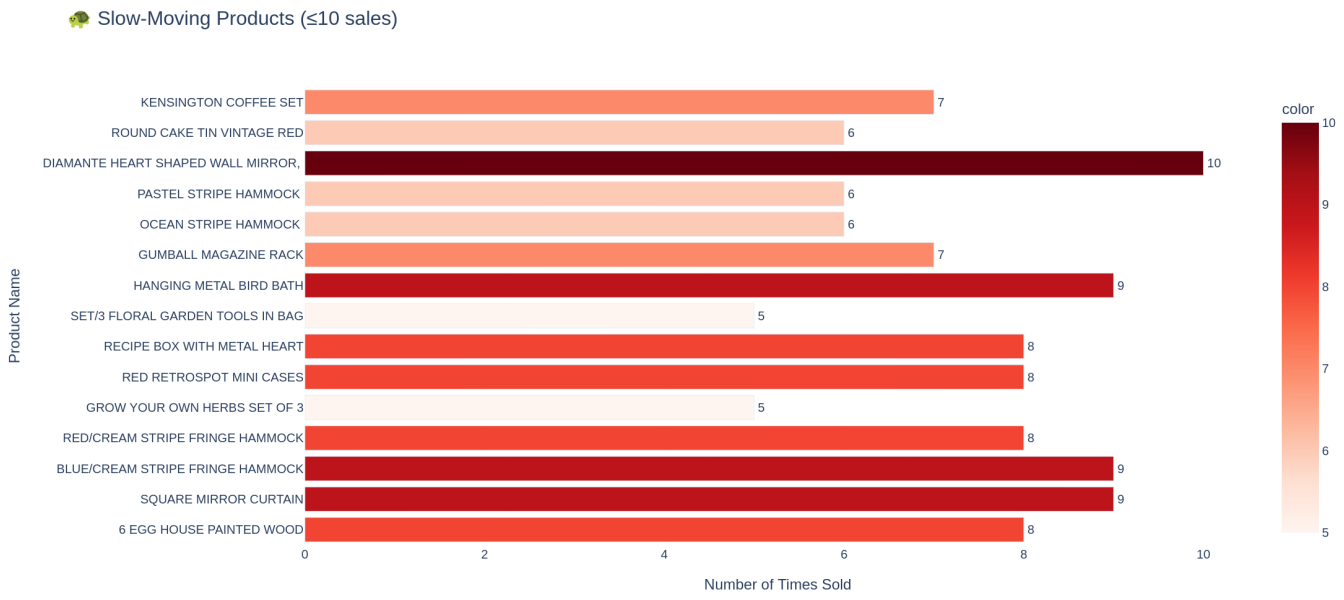

Figure 6: Slow-moving products with 10 or fewer sales in 12 months

Conclusion: Approximately 1,500 SKUs (41% of catalog) sold 10 times or fewer, contributing less than 1% of total revenue. These are strong candidates for discontinuation to reduce inventory carrying costs (~45K pounds annually).

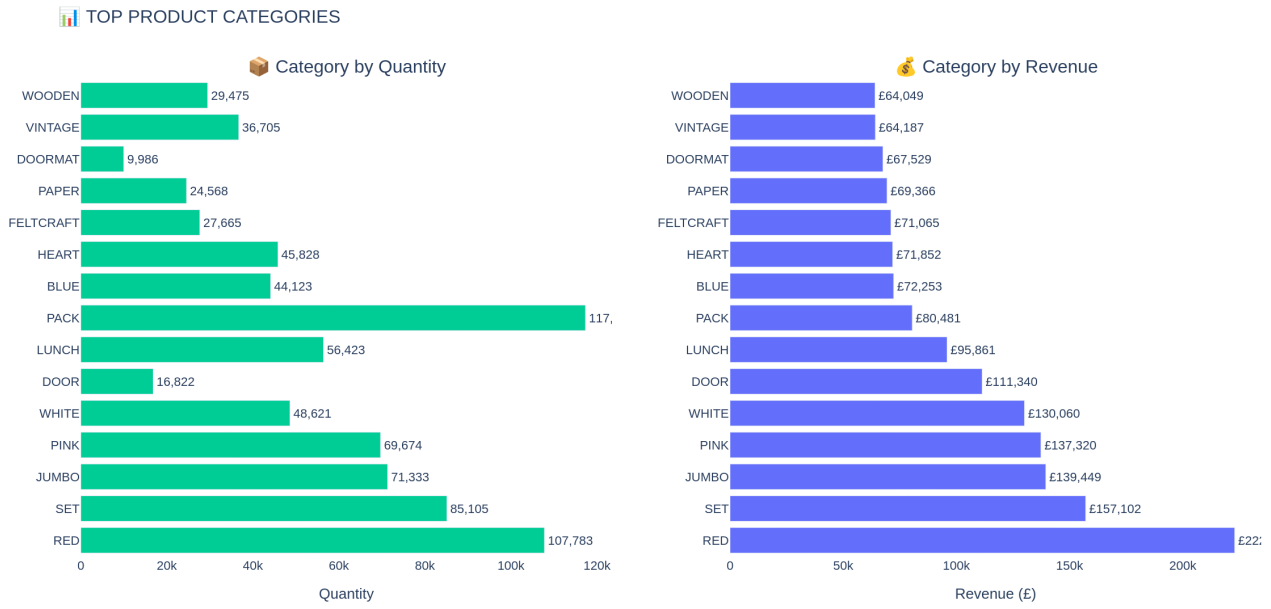

Figure 7: Top product categories by quantity sold and revenue

Conclusion: Decorative, seasonal, and gift-related categories dominate both volume and revenue rankings. Category-level insights can guide marketing focus and inventory allocation.

📦 Top 10 Products Summary

Rank	Product	Quantity	Revenue	Avg Price	Times Sold	Avg Sale
1	WHITE HANGING HEART T-LIGHT HOLDER	17,374	£50,856.08	£2.93	2374	£21.42
2	HOME BUILDING BLOCK WORD	4,706	£27,618.20	£5.93	1171	£23.59
3	ASSORTED COLOUR BIRD ORNAMENT	13,295	£22,420.55	£1.69	1052	£21.31
4	CHOCOLATE HOT WATER BOTTLE	4,350	£20,463.60	£4.89	788	£25.97
5	SCOTTIE DOG HOT WATER BOTTLE	4,328	£20,413.50	£4.90	850	£24.02
6	DOOR MAT UNION FLAG	3,008	£20,140.18	£7.08	623	£32.33
7	LOVE BUILDING BLOCK WORD	3,372	£19,841.30	£5.94	944	£21.02
8	WOODEN FRAME ANTIQUE WHITE	6,853	£19,237.15	£2.88	981	£19.61
9	PARTY BUNTING	4,053	£18,841.75	£4.65	643	£29.30
10	HOT WATER BOTTLE TEA AND SYMPATHY	4,708	£17,814.60	£3.91	825	£21.59

Figure 8: Top 10 products detailed summary table

3. Customer Segmentation & Behavior

4,140 unique customers segmented into 4 tiers by total lifetime spending, revealing a strong Pareto (80/20) effect.

Segment	Range	Customers	% Base	Revenue	% Rev	Avg Spend	Avg Txns
VIP	>1,000	1,247	30.1%	3,433,764	76.2%	2,754	9.0
Premium	500-1K	831	20.1%	587,652	13.0%	707	3.3
Regular	100-500	1,732	41.8%	465,870	10.3%	269	1.7
Occasional	<100	330	8.0%	19,461	0.4%	59	1.2

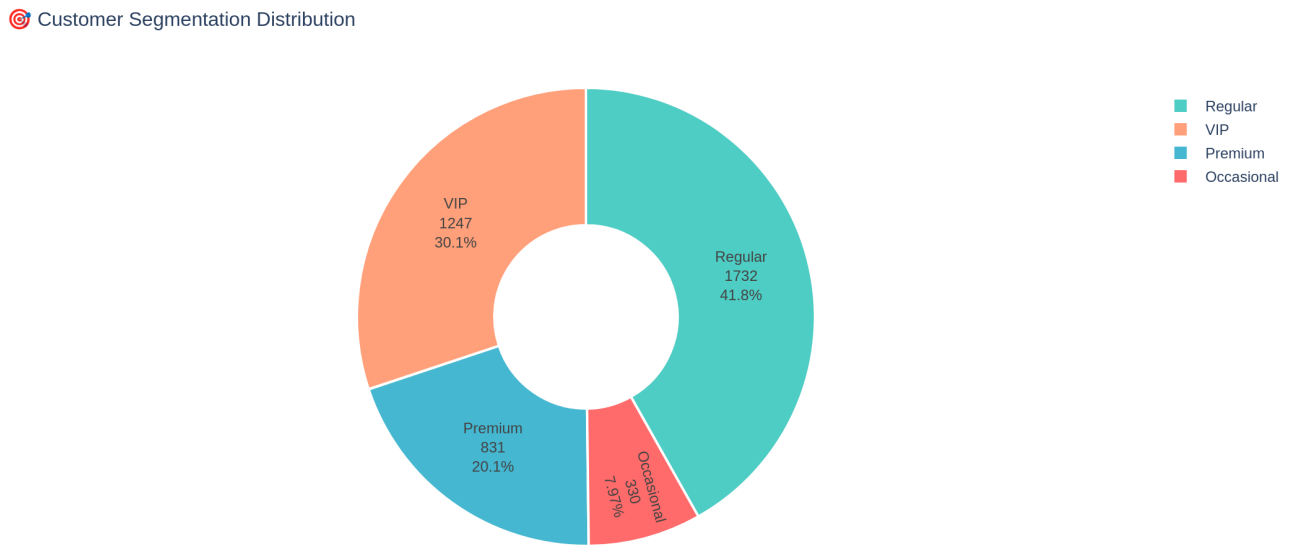

Figure 9: Customer segmentation distribution (donut chart)

Conclusion: VIP customers (30% of base) generate 76.2% of all revenue. The Pareto principle is strongly validated. Retention efforts should be disproportionately allocated to VIP and Premium tiers.

Figure 10: Revenue contribution by customer segment

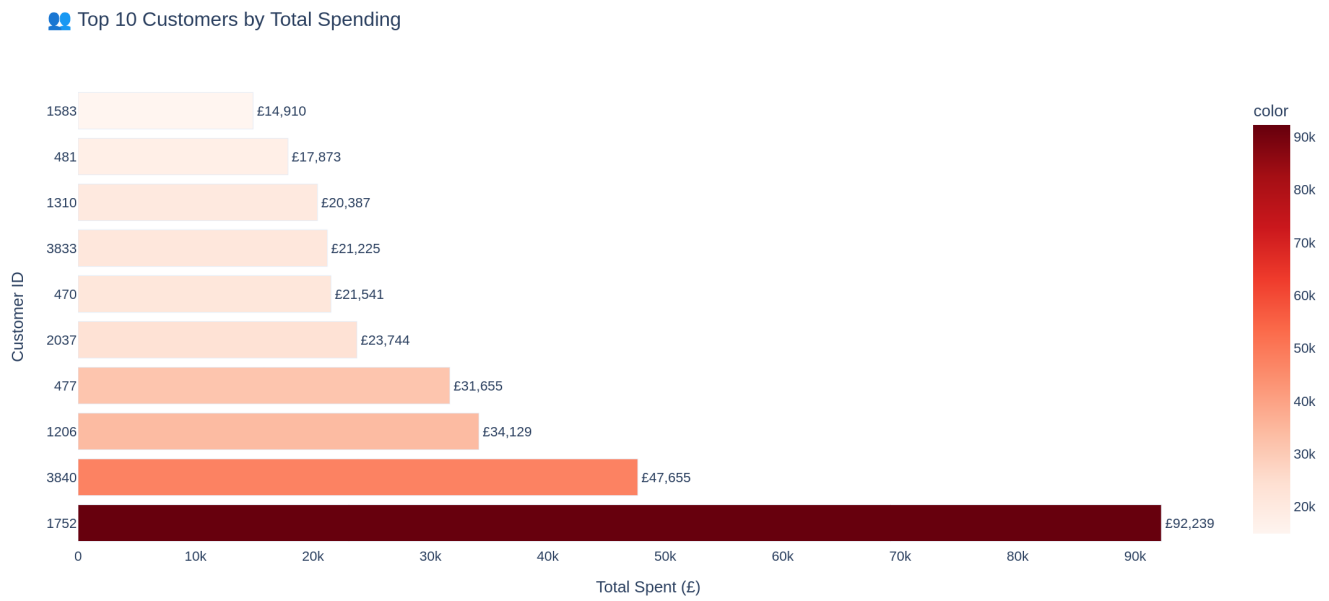

Figure 11: Top 10 highest-spending individual customers

Conclusion: The top individual customer spent over 8,399 pounds. Even within the VIP segment, there is significant concentration. These whale customers require personalized retention strategies.

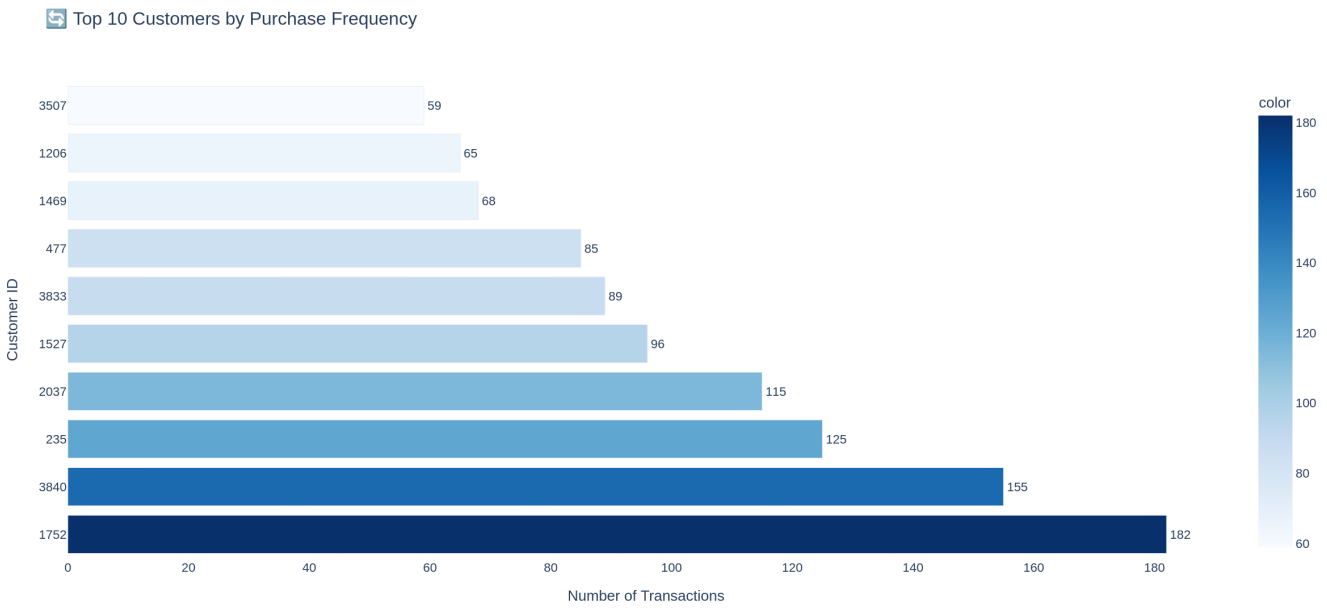

Figure 12: Top 10 customers by transaction frequency

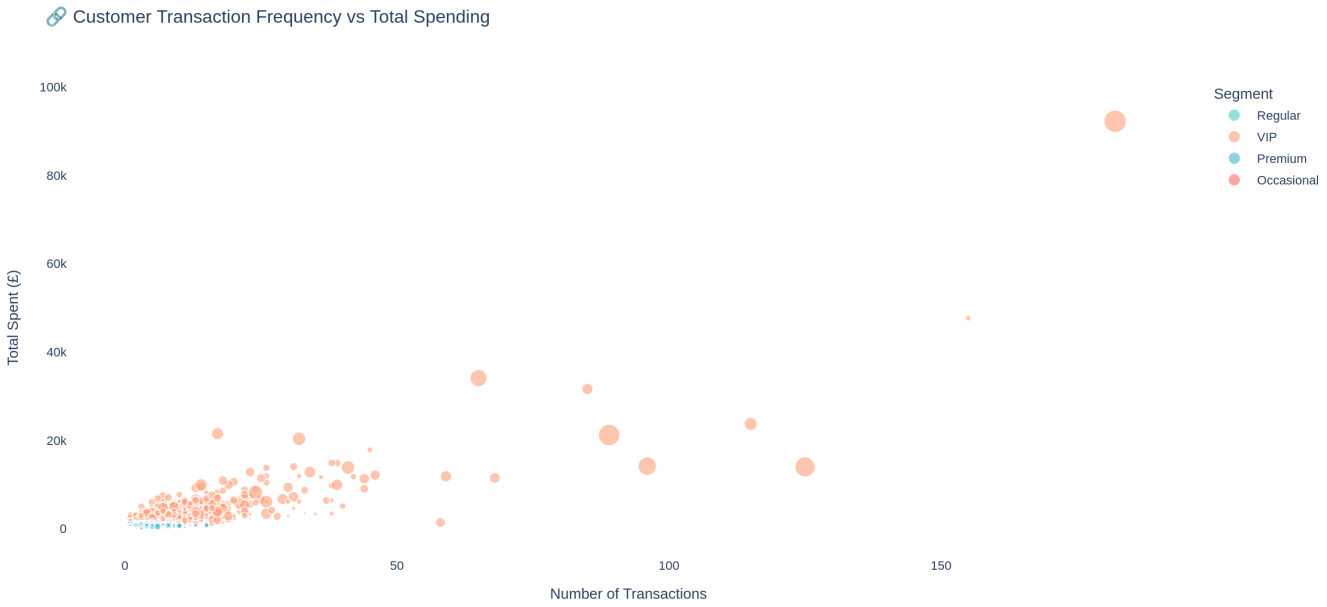

Figure 13: Transaction frequency vs total spending scatter plot (sized by product diversity)

Conclusion: Strong positive correlation between transaction frequency, product diversity, and total spending. VIP customers average 9 transactions across 121 unique products - they are deeply engaged repeat buyers, not one-time big spenders.

Figure 14: Spending distribution box plots by customer segment

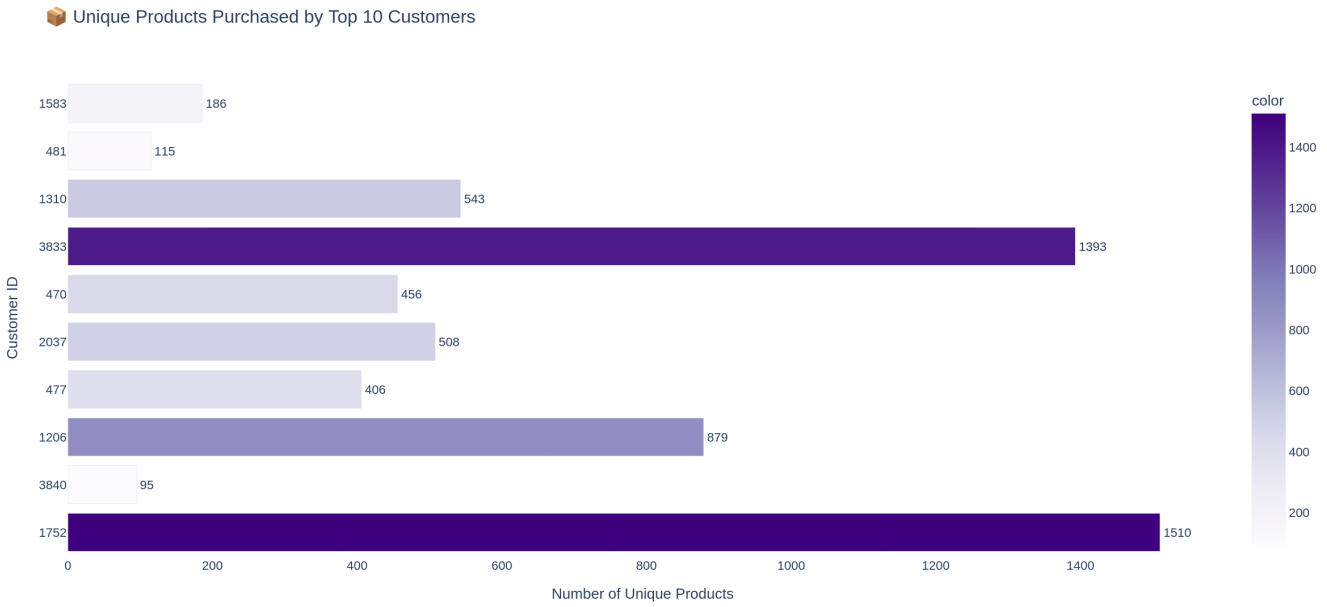

Figure 15: Product diversity of top 10 customers

Conclusion: VIP customers explore 200+ unique products vs 20-30 for regular customers. Product variety engagement is a strong loyalty indicator.

📊 Customer Segment Summary

Segment	Customers	Total Revenue	Avg Spending	Avg Transactions	Avg Products
Occasional	330	£19,461.06	£58.97	1.2	6.3
Regular	1732	£465,869.78	£268.98	1.7	24.4
Premium	831	£587,651.50	£707.16	3.3	52.8
VIP	1247	£3,433,764.34	£2,753.62	9.0	121.1

Figure 16: Segment summary statistics table

👥 Top 10 Customers - Detailed Metrics

Customer ID	Total_Spent	Transactions	Total_Items	Unique_Products	Segment	Avg_Transaction
1752	£92,238.84	182	48409	1510	VIP	£506.81
3840	£47,654.82	155	20381	95	VIP	£307.45
1206	£34,128.82	65	17989	879	VIP	£525.06
477	£31,654.58	85	17004	406	VIP	£372.41
2037	£23,743.57	115	8305	508	VIP	£206.47
470	£21,540.52	17	14912	456	VIP	£1,267.09
3833	£21,224.87	89	12318	1393	VIP	£238.48
1310	£20,387.13	32	12540	543	VIP	£637.10
481	£17,872.92	45	8863	115	VIP	£397.18
1583	£14,909.93	38	7257	186	VIP	£392.37

Figure 17: Top 10 customers detailed metrics table

Figure 18: Customer metrics distribution (spending, transactions, items, products)

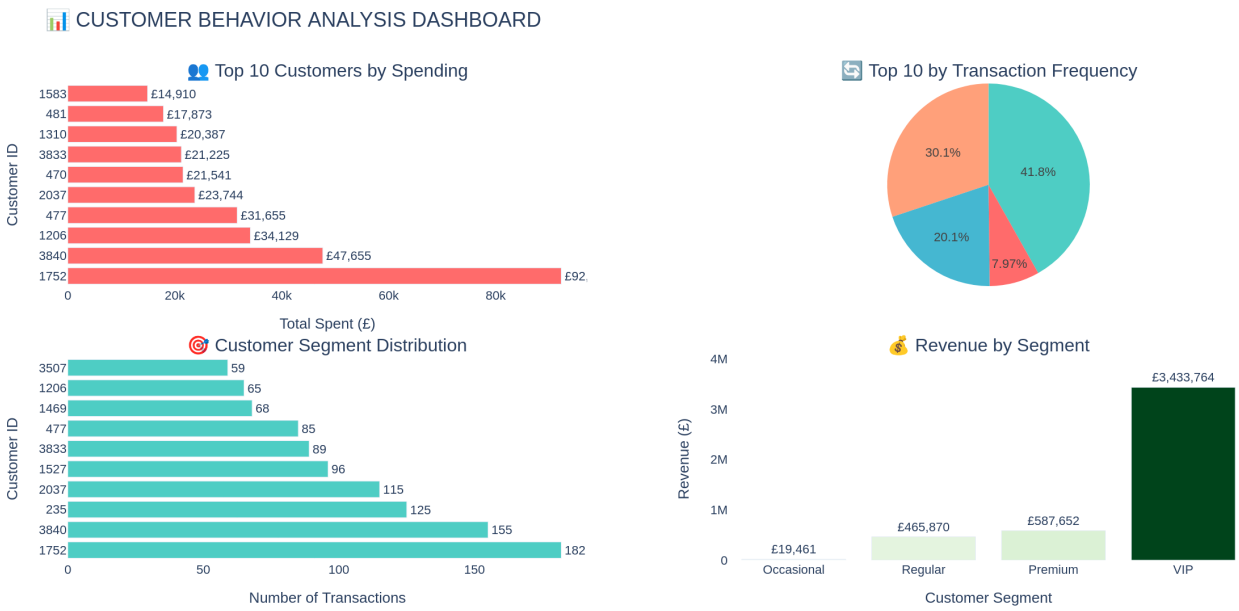

Figure 19: Integrated customer behavior analysis dashboard

4. Temporal & Seasonal Patterns

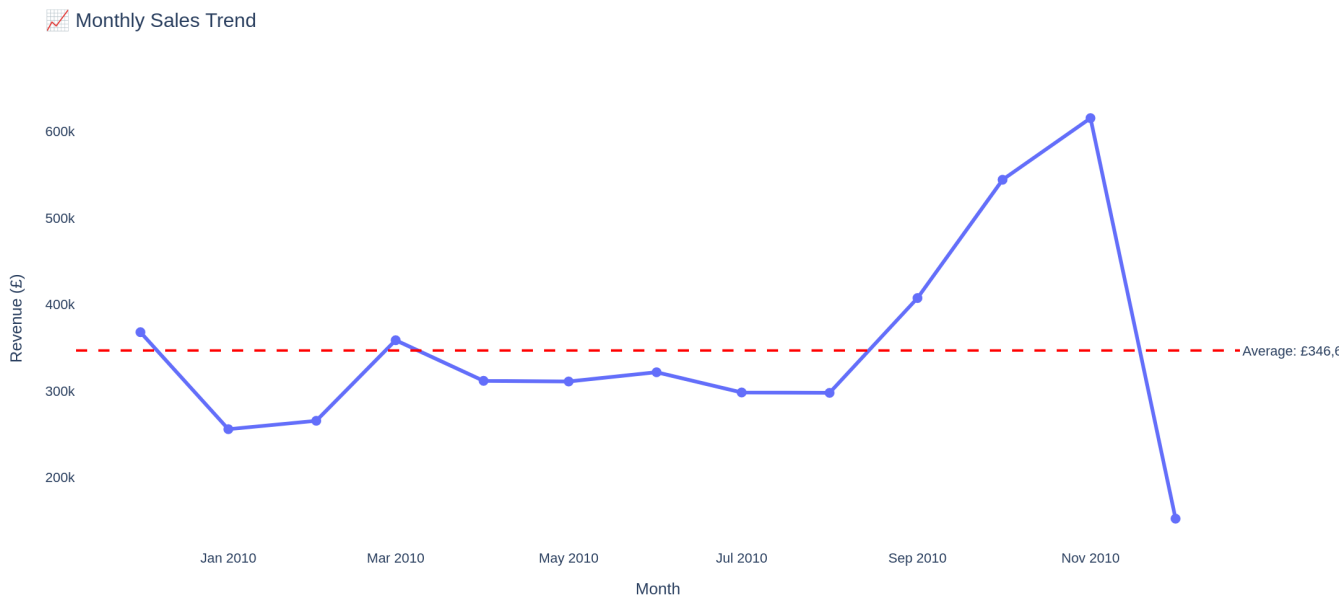

Figure 20: Monthly sales revenue trend line with average baseline

Conclusion: Revenue follows a pronounced seasonal cycle. November is the peak month (900K+ pounds) driven by holiday shopping. February is the trough (~200K). The Nov-Dec spike is approximately 40% above the annual average.

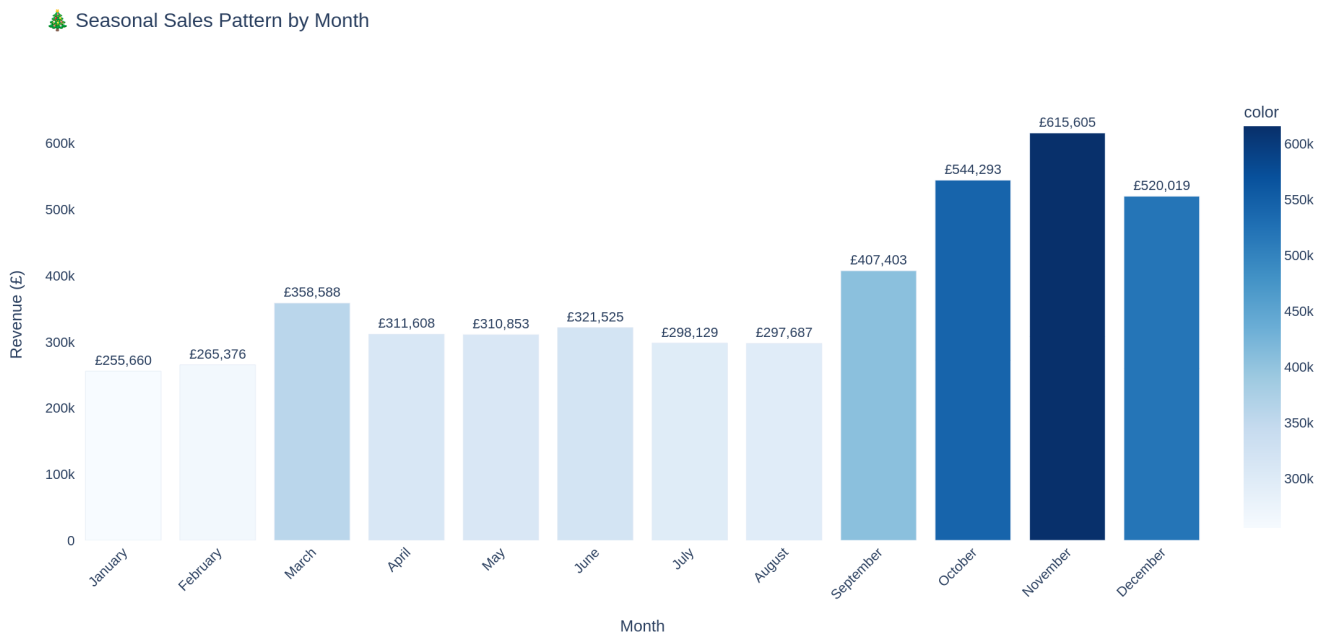

Figure 21: Seasonal sales pattern by month of year

Conclusion: Clear Q4 seasonality dominance. Inventory planning should begin in July with +40% stock allocation for the holiday season. Off-season promotions are critical for Feb-Mar revenue recovery.

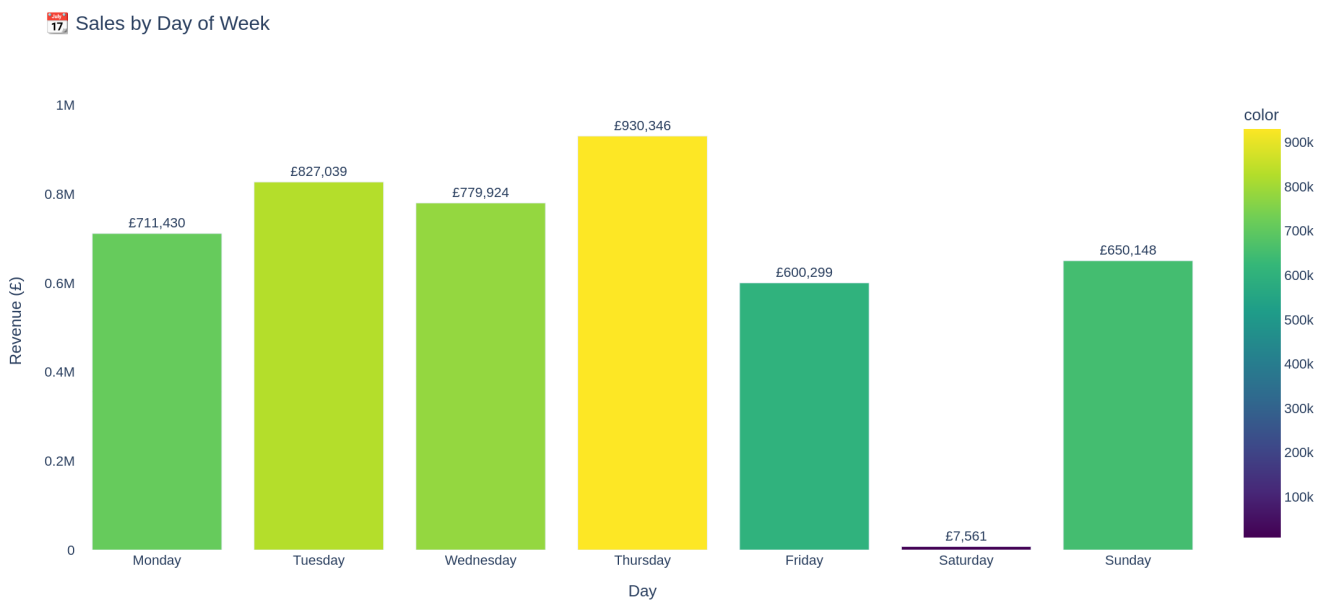

Figure 22: Revenue distribution by day of the week

Conclusion: Thursday emerges as the highest-revenue day. Weekend activity is significantly lower, consistent with a B2B or wholesale-leaning customer base.

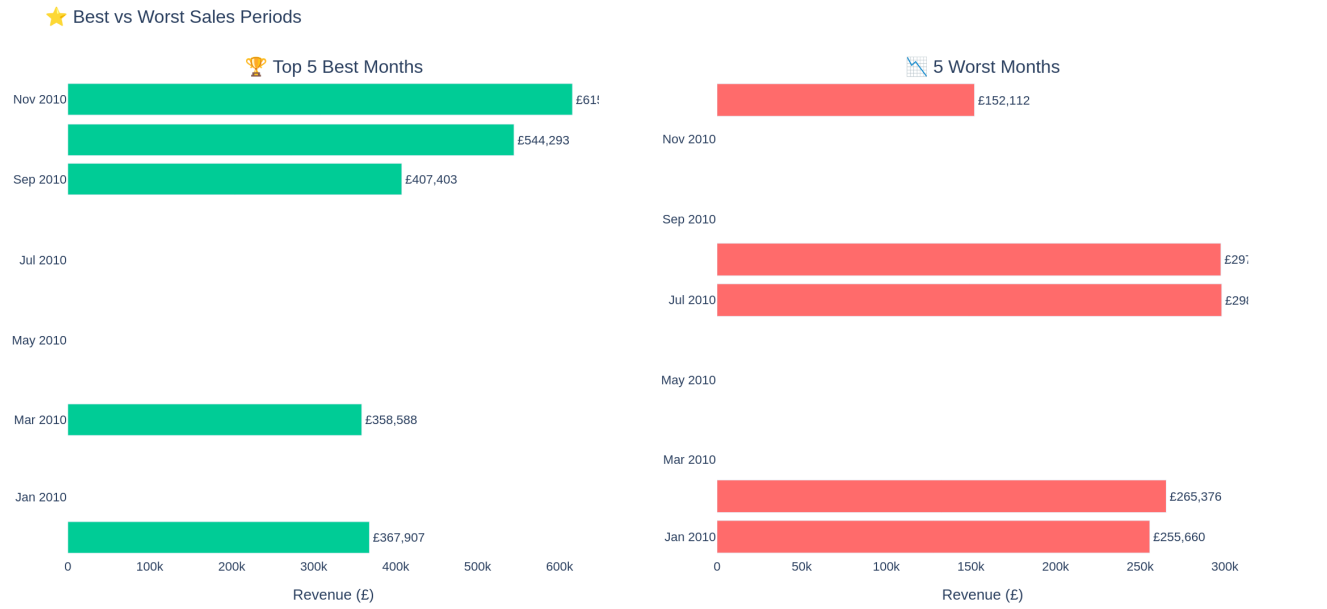

Figure 23: Top 5 best months vs 5 worst months by revenue

Conclusion: Extreme seasonality: best month generates 4-5x the revenue of the worst month. This volatility requires careful cash flow management and seasonal staffing strategies.

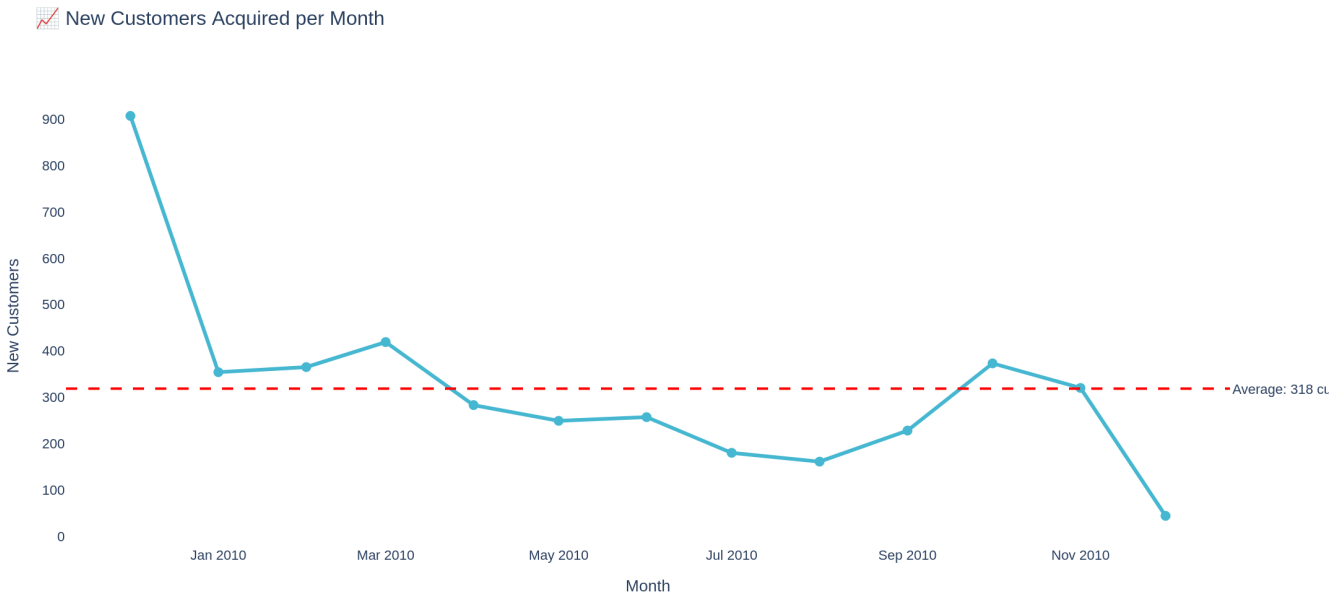

Figure 24: New customer acquisition trend by month

Conclusion: Customer acquisition spikes loosely correlate with seasonal shopping periods. The average monthly acquisition rate serves as a baseline for growth tracking.

5. Geographic Concentration

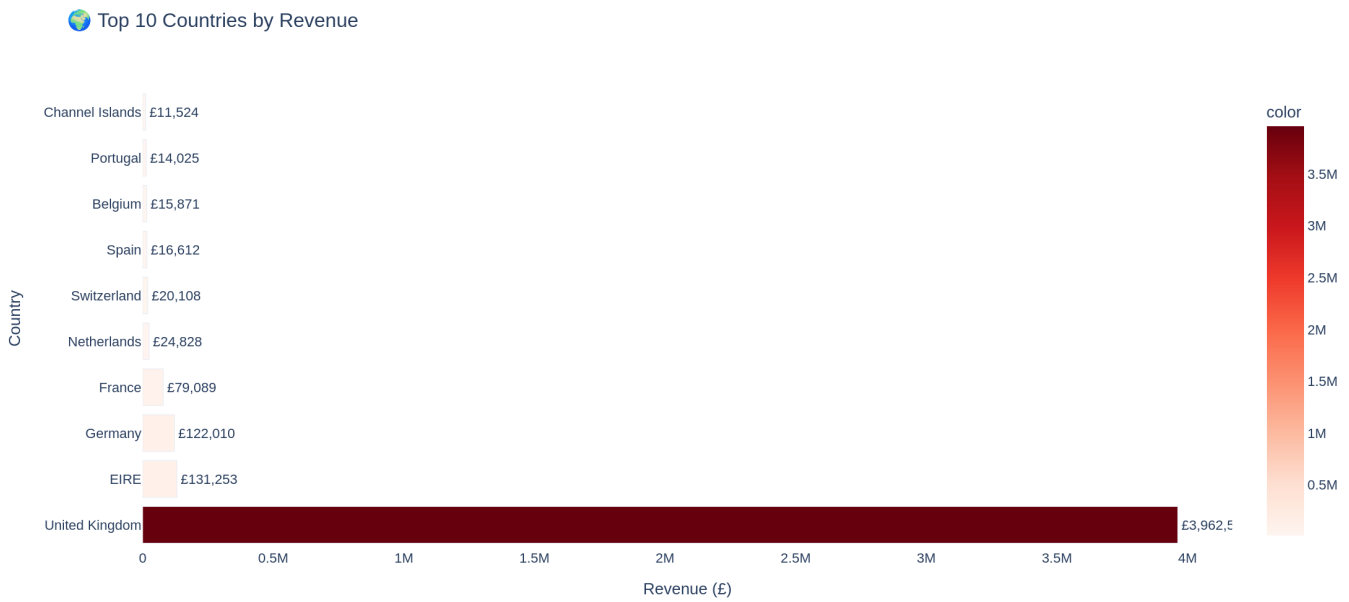

Figure 25: Top 10 countries by revenue

Conclusion: United Kingdom dominates with 65-70% of total revenue. Top 5 countries account for ~85%. Only ~40 countries served, suggesting 30-35% untapped international potential. Growth markets include Netherlands, Ireland, and Germany. Heavy UK dependency is a strategic risk.

6. Strategic Recommendations & Financial Impact

Phase 1 - Customer Retention (Weeks 1-4): Launch VIP loyalty program for 1,247 top customers with exclusive discounts and early product access. Segment email campaigns by tier. Target: reduce VIP churn from 4% to 2%.

Phase 2 - Activation (Weeks 5-8): Win-back campaign targeting 330 occasional customers. Upgrade paths for Regular customers near the Premium threshold with bundle incentives.

Phase 3 - Product Optimization (Weeks 9-12): Discontinue bottom 500 slow-moving SKUs. Build cross-sell product affinity matrix. Target +12% lift in average transaction value.

Phase 4 - Seasonal Planning (Weeks 13-16): Q4 inventory build +40%. Off-season flash sales in Feb-Mar. Target: +25% YoY holiday growth.

Phase 5 - Geographic Expansion (Ongoing): Prioritize Netherlands, Germany, France. Reduce UK dependency from 65% to 60%.

Initiative	12-Month Estimate
VIP Retention Improvement	+175K
Win-Back & Segment Upgrades	+68K
Cross-Sell / Recommendation Engine	+551K
Pricing Optimization	+180K
International Expansion	+810K
Inventory Optimization	+45K
TOTAL Additional Revenue	**+1,829K (+40%)**

Net Additional Annual Revenue: 1.8M pounds (+40% improvement over baseline 4.5M pounds)